

One other note I want to make is that if $\langle x_\alpha \rangle_{\alpha \in A}$ is an ultra-net and $X = \bigcup_{i=1}^n E_i$, then we can conclude that there is at least one E_i such that x_α is eventually in E_i .

Proof:

We proceed by induction on n . For our base case, suppose $X = E_1 \cup E_2$. Then if x_α is not eventually in E_1 , we know that x_α is eventually in $X - E_1$. And since $X - E_1 \subseteq E_2$, we can conclude that x_α is eventually in E_2 .

Now suppose $n > 2$ and that $X = \bigcup_{i=1}^n E_i$. Then by writing:

$$X = (E_1 \cup E_2) \cup E_3 \cup \dots \cup E_n,$$

we can conclude by induction that either there exists $i \geq 3$ with x_α eventually in E_i , or we have that x_α is eventually in $E_1 \cup E_2$. We're trivially done in the former case. So assume the latter case is what happens.

Next, write $X = E_1 \cup (E_2 \cup E_3) \cup E_4 \cup \dots \cup E_n$. Then we know from our induction hypothesis that either there exists $i \neq 2, 3$ with x_α eventually in E_i or we have that x_α is eventually in $E_2 \cup E_3$. We assume the former case doesn't happen since otherwise we'd be trivially done. In particular, this means that x_α is not eventually in E_1 .

But now we have that x_α is eventually in $(E_1 \cup E_2)$, and yet at the same time since $\langle x_\alpha \rangle_{\alpha \in A}$ is an ultrafilter, we can conclude that x_α is eventually in $X - E_1$. In turn, we can conclude that x_α is eventually in $(E_1 \cup E_2) \cap (X - E_1) = E_2 - E_1 \subseteq E_2$. ■

Proposition: Suppose X is a uniform space. Then X is totally bounded if and only if every net $\langle x_\alpha \rangle_{\alpha \in A}$ in X admits a Cauchy subnet.

($\Rightarrow$)

Let $\langle x_{\alpha_\beta} \rangle_{\beta \in B}$ be an ultra-subnet of $\langle x_\alpha \rangle_{\alpha \in A}$. Then I claim that $\langle x_{\alpha_\beta} \rangle_{\beta \in B}$ is Cauchy since more generally, any ultra-net $\langle y_\gamma \rangle_{\gamma \in C}$ is Cauchy in a totally bounded space.

To see why, let $\{y_\gamma\}_{\gamma \in C}$ be an ultra-net. Then suppose E is any entourage of X . Firstly, we find an entourage E' such that $E' = (E')^{-1}$ and $E' \circ E' \subseteq E$. Next, note that since X is totally bounded, there exists $x_1, \dots, x_n \in X$ such that $X = \bigcup_{i=1}^n E'[x_i]$. Since $\{y_\gamma\}_{\gamma \in C}$ is an ultra-net, we know there must exist some x_i such that y_γ is eventually in $E'[x_i]$.

But now let $\gamma_0 \in C$ be such that $y_\gamma \in E'[x_i]$ for all $\gamma \geq \gamma_0$. Then for all $\gamma_1, \gamma_2 \geq \gamma_0$ in C we have that $x_i E' y_{\gamma_1}$ and $x_i E' y_{\gamma_2}$. This implies that $y_{\gamma_1} E y_{\gamma_2}$.

($\Leftarrow$)

Suppose that X is not a totally bounded space. Thus, we know there exists an entourage E such that $E[S]$ is a proper subset of X for all finite subsets S of X .

Let $\mathcal{S}$ be the collection of all finite subsets of X . This becomes a directed set when partially ordered by inclusion. Next, for every $S \in \mathcal{S}$ let x_S be an element of $X - E[S]$. Then, we can consider the net $\langle x_S \rangle_{S \in \mathcal{S}}$.

I claim that $\langle x_S \rangle_{S \in \mathcal{S}}$ has no Cauchy subnets. To see why, suppose that $\langle x_{S_\beta} \rangle_{\beta \in B}$ is any subnet of $\langle x_S \rangle_{S \in \mathcal{S}}$. Then consider any fixed $\beta_0 \in B$. It suffices to show that there exists $\beta_1, \beta_2 \geq \beta_0$ in B such that $(x_{S_{\beta_1}}, x_{S_{\beta_2}}) \notin E$. Fortunately, just let $\beta_1 = \beta_0$ and then let $\beta_2 \in B$ be such that $\beta_2 \geq \beta_0$ and $S_{\beta_2} \supseteq S_{\beta_0} \cup \{x_{S_{\beta_0}}\}$. Then we know $(x_{S_{\beta_1}}, x_{S_{\beta_2}}) \notin E$. ■

Before wrapping up for the day, I also want to translate the prior result into a result about Cauchy sequences. I think this is especially worth doing since the constructions in this simpler case are less set-theoretic.

Proposition: Suppose X is a uniform space whose uniform structure is generated by a countable filterbase $\{E_k\}_{k \in \mathbb{N}}$. Then X being totally bounded implies every sequence $\{x_n\}_{n \in \mathbb{N}}$ in X admits a Cauchy subsequence.

Proof:

For each k we can pick E'_k such that $E'_k = (E'_k)^{-1}$ and $E'_k \circ E'_k \subseteq E_k$. Then because X is totally bounded, we know for each k that there exists $y_1^{(k)}, \dots, y_{m(k)}^{(k)} \in X$ such that $X = \bigcup_{j=1}^{m(k)} E'_k[y_j^{(k)}]$.

But now suppose $\{x_n\}_{n \in \mathbb{N}}$ is any sequence in X . Then there exists some $y_j^{(1)}$ as well as some subsequence $\{x_{n_i^{(1)}}\}_{i \in \mathbb{N}}$ such that $x_{n_i^{(1)}} \in E'_1[y_j^{(1)}]$ for all i . Next, there exists a subsequence $\{x_{n_i^{(2)}}\}_{i \in \mathbb{N}}$ of $\{x_{n_i^{(1)}}\}_{i \in \mathbb{N}}$ as well as some $y_j^{(2)}$ with $x_{n_i^{(2)}} \in E'_2[y_j^{(2)}]$ for all i . Continuing on in this manner, for every $k \in \mathbb{N}$ we can find a sequence $\{x_{n_i^{(k)}}\}_{i \in \mathbb{N}}$ which is a subsequence of $\{x_{n_i^{(k')}}\}_{i \in \mathbb{N}}$ for all $k' < k$ and which also satisfies that there exists $y^{(k)} \in X$ with $x_{n_i^{(k)}} \in E'_k[y^{(k)}]$ for all i .

To finish off, we now do a diagonal argument on all of the previous sequences we constructed. Namely, consider the sequence $\{z_k\}_{k \in \mathbb{N}}$ where $z_k := x_{n_k^{(k)}}$. This is a subsequence of $\{x_n\}_{n \in \mathbb{N}}$. Also I claim it is Cauchy.

After all, suppose D is an entourage of X . Then we can find an entourage E_k such that $E_k \subseteq D$. Next, note for all $k_1, k_2 \geq k$ that $y^{(k)} E'_k z_{k_1}$ and $y^{(k)} E'_k z_{k_2}$. Hence, we know that $z_{k_1} E_k z_{k_2}$. So, $(z_{k_1}, z_{k_2}) \in E_k \subseteq D$ for all $k_1, k_2 \geq k$. ■

Proposition: Suppose X is a uniform space. If every sequence $\{x_n\}_{n \in \mathbb{N}}$ in X admits a Cauchy subsequence then X is totally bounded.

Proof:

Suppose X is not totally bounded and let E be an entourage such that $E[S] \neq X$ for all finite subsets S of X . By shrinking E , we can without loss of generality assume that E is symmetric (i.e. $E = E^{-1}$). Now pick any $x_1 \in X$. We can find $x_2 \in X - E[x_1]$. Next, we can find $x_3 \in X - E[x_1, x_2]$. And continuing on in this fashion, we can find $x_n \in X - \bigcup_{i=1}^{n-1} E[x_i]$. Then we consider the sequence $\{x_n\}_{n \in \mathbb{N}}$.

Trivially, no subsequence (or even any subnet I now realize) of $\{x_n\}_{n \in \mathbb{N}}$ will be Cauchy since $(x_{n_1}, x_{n_2}) \notin E$ whenever $n_1 \neq n_2$. ■

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Today I want to start tying the theory of uniform spaces to the theory of topology. So, I'll continue loosely following the book I was following before.

Let $(X, \mathcal{U})$ be a uniform space. Then taking inspiration from [pages 798-800](#) as well as from how the metric space topology is defined, we'll define that a set $U \subseteq X$ is open in the uniform topology if for all $x \in U$ there exists an entourage $E \in \mathcal{U}$ with $E[x] \subseteq U$.

Proposition: The uniform topology is a well-defined topology.

Proof:

To start off, we trivially have that $\emptyset$ and X are open sets in the uniform topology.

Next suppose U_1 and U_2 are open sets in the uniform topology, and consider any $x \in U_1 \cap U_2$. Since U_1 and U_2 are open, we can find entourages E_1 and E_2 such that $E_1[x] \subseteq U_1$ and $E_2[x] \subseteq U_2$. So, we then consider the entourage $E := E_1 \cap E_2$. Now $E[x] \subseteq U_1 \cap U_2$. And since x was arbitrary, we've proven that $U_1 \cap U_2$ is an open set in the uniform topology.

(By induction we can in turn conclude that any finite intersection of open sets is open in the uniform topology.)

Finally, suppose $\{U_\alpha\}_{\alpha \in A}$ is an arbitrary collection of open sets in the uniform topology. Then for any $x \in \bigcup_{\alpha \in A} U_\alpha$ we know that there exists $\alpha_0 \in A$ with $x \in U_{\alpha_0}$. Next, since U_{α_0} is open we know that there exist an entourage E such that $E[x] \subseteq U_{\alpha_0}$. And now $E[x] \subseteq \bigcup_{\alpha \in A} U_\alpha$. This proves that $\bigcup_{\alpha \in A} U_\alpha$ is open. ■

Proposition: Suppose $\mathcal{B}$ is a filterbase for $\mathcal{U}$. Then for any $x \in X$ we have that $\{E[x] : E \in \mathcal{B}\}$ is a local base of neighborhoods of x . (See [page 796](#) for the definition of a local base of neighborhoods.)

Proof:

To start off, suppose N is any neighborhood of x . Then we know there is an open set U such that $x \in U \subseteq N$. But now since U is open, we know there exists some entourage $D \in \mathcal{U}$ such that $D[x] \subseteq U$. And finally, since $\mathcal{B}$ is a filterbase for $\mathcal{U}$ we know that there exists another entourage $E \in \mathcal{B}$ with $E \subseteq D$. In turn, $E[x] \subseteq D[x]$. And with that, we've proven that for all neighborhoods N of x there exists $E \in \mathcal{B}$ with $x \in E[x] \subseteq N$.

Meanwhile, we also need to show that $E[x]$ is a neighborhood of x for all $E \in \mathcal{B}$. So, having fixed our entourage E , let $U := \{y \in X : \exists D \in \mathcal{U} \text{ s.t. } D[y] \subseteq E[x]\}$. Note that:

- $x \in U$ since $E \in \mathcal{U}$ and $E[x] \subseteq E[x]$.
- Also since $y \in D[y]$ for all entourages $D \in \mathcal{U}$ and $y \in X$, we know that $U \subseteq E[x]$.
- Finally, U is open. After all, consider any $y \in U$ and let $D \in \mathcal{B}$ be such that $D[y] \subseteq E[x]$. But now since $\mathcal{U}$ is a uniform structure, we can find some $D' \in \mathcal{U}$ such that $D' \circ D' \subseteq D$. And now I claim that $D'[y] \subseteq U$.

After all, suppose $z \in D'[y]$. Then for any $w \in D'[z]$ we have that $yD'z$ and $zD'w$. It follows that $yD' \circ D'w$. So, $w \in D[y] \subseteq E[x]$. And this proves that

$D'[z] \subseteq E[x]$. In turn, by the definition of U we know that $z \in U$. And since z was arbitrary, this proves that $D'[y] \subseteq U$.

With that, we can conclude that $E[x]$ is a neighborhood of x for every entourage $E \in \mathcal{B}$. ■

Corollary: If (X, ρ) is a metric space, then the uniform topology associated with the metric uniform structure is precisely the standard metric topology.

Proof:

Let $\{D_\varepsilon\}_{\varepsilon>0}$ be the standard uniformity base for the metric uniform structure. Then we can see for all $x \in X$ that $D_\varepsilon[x]$ is just the ball of radius ε centered at x . Hence, for any $x \in X$ we have that $\{D_\varepsilon[x], \varepsilon > 0\}$ is a local base of neighborhoods of x in both the standard metric topology and in the uniform topology. It follows that convergence in both topologies are equivalent, and in turn the topologies are equal to each other. ■

Corollary 2: Let G be a topological group. Then the uniform topologies associated with the left uniform structure and the right uniform structure are both equivalent to the original topology on G .

Proof:

Just show like in the prior corollary that either of the uniform topologies on G admits a local base of neighborhoods at any $x \in G$ which is also a local base of neighborhoods of x in the original topology. After all given any neighborhood N of the identity element $e \in G$ we have that $L_N[x] = xN$ and $R_N[x] = Nx$.

As a side note, since the left and right uniform structures on a topological group are generally not the same, this corollary also demonstrates that while we can speak of a single canonical topology associated with a uniform space, we generally can't speak of a single canonical uniform structure whose associated topology is some desired topology.

There are two properties of the uniform topology which had better be true. Namely, uniform continuity should be stronger than continuity, and also every convergent net should be Cauchy. I'll prove these facts below.

Proposition: Suppose X and Y are uniform spaces and $\phi : X \rightarrow Y$ is uniformly continuous. Then ϕ is continuous when X and Y are equipped with the uniform topologies.

Proof:

Consider any $x \in X$ and then consider any neighborhood N of $\phi(x)$ in Y . We know there exists an entourage E of Y with $E[\phi(x)] \subseteq N$. Yet also note that $D := (\phi \times \phi)^{-1}(E)$ is an entourage of X since ϕ is uniformly continuous. And now I claim that $D[x]$ is a neighborhood of $x \in X$ satisfying that $\phi(D[x]) \subseteq N$. After all, if $y \in D[x]$ then $\phi(y) \in E[\phi(x)] \subseteq N$. This proves that ϕ is continuous at x . And since x was arbitrary, we can conclude that ϕ is continuous everywhere. ■

Proposition: Suppose X is a uniform space and that the net $\langle x_\alpha \rangle_{\alpha \in A}$ converges with respect to the uniform topology to a point $x \in X$. Then $\langle x_\alpha \rangle_{\alpha \in A}$ is a Cauchy net.

Proof:

Let E be any entourage of X and then pick a symmetric entourage E' of X such that $E' \circ E' \subseteq E$. Since $x_\alpha \rightarrow x$ in the uniform topology, we know that the x_α are eventually in $E'[x]$. But now let $\alpha_0 \in A$ satisfy that $x_\alpha \in E'[x]$ for all $\alpha \geq \alpha_0$. Then for all $\alpha, \beta \geq \alpha_0$ we have that $x_\alpha E' x_\alpha$ and $x_\alpha E' x_\beta$ implies that $x_\alpha E' \circ E' x_\beta$. So, $x_\alpha E x_\beta$ for all $\alpha, \beta \geq \alpha_0$. And this proves that our net is Cauchy. ■

Next I want to prove another useful fact about uniform the uniform topology.

Proposition: Suppose X is a uniform space. Then the following are equivalent:

- (a) X is a separated uniform space,
- (b) the uniform topology on X is T_1 ,
- (c) the uniform topology on X is Hausdorff.

Proof: